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Linear Algebra

⇒ It's the branch of mathematics that deals with vectors & matrices.

⇒ Vector $\rightarrow$ 1D array

Matrix $\rightarrow$ 2D array

⇒ Numpy, PyTorch are fully vectorized, hence these operations be 100-1000 times faster in execution.

⇒ LA appⁿ in ML:

① In ML, structured data is represented as matrices and vectors.

② Unstructured data like text can also be represented as a vector in NLP.

③ Images & videos ^{& audios} can be represented as matrix & tensor.

④ In ML model training & prediction, it involves sequence of matrix & vector multiplication.

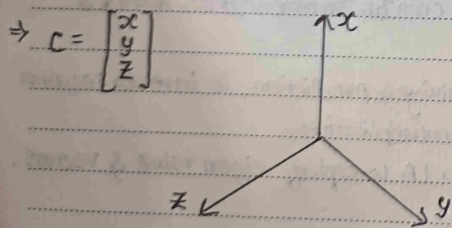
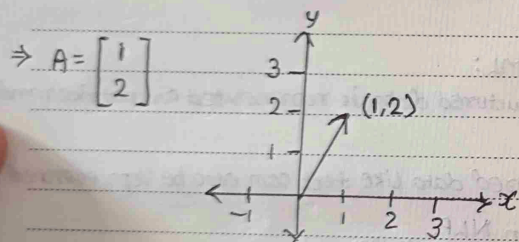
⑤ In PCA, we use LA concept of eigen value & vectors.

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Scalars and Vectors:

⇒ **Scalar:** It is a single numerical value. It represents a magnitude or quantity and has no direction.

⇒ **Vector:** Numerical value which has both magnitude & dirⁿ.
A vector is an ordered list of numbers. It can represent a point in space or quantity with both magnitude & dirⁿ.
→ collection of values



(एक है) ⇒ **unit vector:** magnitude 1 (one)

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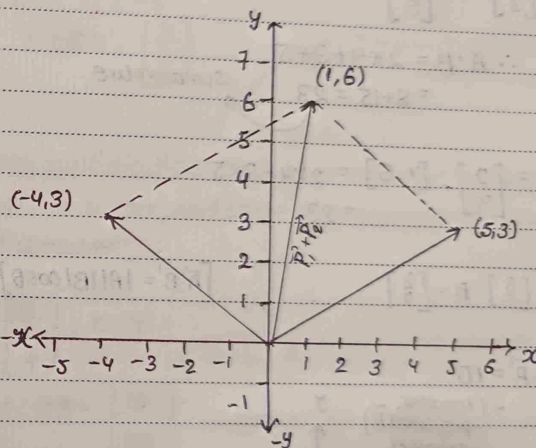
$$(3, 3) \Rightarrow 3\hat{i} + 3\hat{j}$$

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Addⁿ of 2 vectors:

$$\Rightarrow P_1 = \begin{bmatrix} -4 \\ 3 \end{bmatrix} \quad P_2 = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\therefore P_1 + P_2 = \begin{bmatrix} -4 \\ 3 \end{bmatrix} + \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$



Multiplication of Vectors:

- ⇒ 3 types: ① dot product (inner product)
② Element wise multⁿ
③ Scalar multⁿ

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used in: ① Cosine similarity
② Recommendation system
③ Vector databases

⇒ Dot Product: of 2 vectors results in a scalar and is calculated as the sum of the products of their corresponding components.

e.g.,

$$A = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, B = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$\therefore A \cdot B = 2 \times 4 + 3 \times 5 = 8 + 15 = 23 \quad \text{scalar value}$$

Also,

$$A \cdot B^T = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 & 5 \end{bmatrix} = 2 \times 4 + 3 \times 5 = 23$$

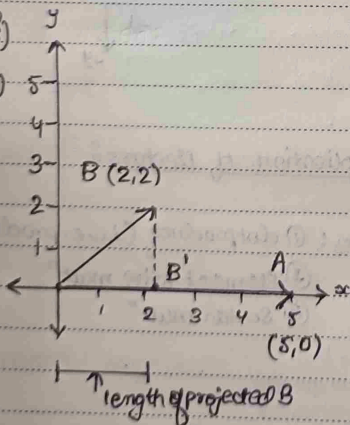
Also,

$$A = \begin{bmatrix} 5 \\ 0 \end{bmatrix}, B = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad [A \cdot B = |A||B|\cos\theta]$$

$$\therefore \vec{A} \cdot \vec{B} = 10$$

= (length of projected B) $\times$ (length of vector A)

$$= 2 \cdot 5 = 10$$



→ appⁿ: ① Feature Engineering
② RNN, LSTM RNN

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⇒ Element wise multⁿ: In this, corresponding elements of 2 vectors are multiplied to form a new vector of the same dimension.

e.g.,

$$A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, B = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$$

$$\therefore A \odot B = \begin{bmatrix} 3 \\ 8 \\ 15 \end{bmatrix}$$

appⁿ: ① Normalization & standardization

⇒ Scalar multiplication: Multiplying vector by a scalar, where result is a vector, and where each component is scaled by the vector.

e.g.,

$$A = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}, C = 4$$

$$\therefore CA = \begin{bmatrix} 12 \\ 20 \\ 28 \end{bmatrix}$$

- applications:
- ① Data representation
 - ② Images in Computer vision
 - ③ Confusion matrix
 - ④ Neural Network

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Matrices:

⇒ It's a rectangular array of nos., symbols or expressions arranged in rows & columns.
e.g.,

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots \\ a_{21} & a_{22} & \dots \\ \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots a_{mn} \end{bmatrix}_{m \times n}$$

row column

Matrices Operations:

⇒ They provide equation to manipulate and analyse multidimensional data efficiently.

⇒ Matrix addⁿ and subtraction:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \\ 1 & 2 & 3 \end{bmatrix}$$

$$\therefore A+B = \begin{bmatrix} 5 & 7 & 9 \\ 11 & 13 & 15 \\ 8 & 10 & 12 \end{bmatrix}$$

→ appⁿ: adjusting the value by of some external factors.

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⇒ Scalar multiplication: multiply the element (scalar) with every element of the matrix.
e.g.,

$$P = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \therefore C = 2$$

$$\therefore P_{\text{adjusted}} = CP = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix}$$

⇒ Matrix Multiplication: It involves the dot product of rows of the first matrix with columns of the second matrix.

$$A_{m \times n} \cdot B_{n \times p} = C_{m \times p}$$

e.g.,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad B = \begin{bmatrix} 7 & 9 & 11 \\ 8 & 10 & 12 \end{bmatrix}$$

$$\therefore C = A \cdot B = A \cdot B^T$$

$$= \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}$$